Course: C Programming

From Fundamentals to Problem Solving

Duration

- 45 Hours (Standard Course)
- 60 Hours (With Lab Practice)
- Suitable for:
    - Intermediate Students
    - B. Tech 1st Year Students
    - Diploma Students
    - Beginners in Programming

Module 1: Introduction to Computers & Programming

Topics

- Introduction to Computers
- Hardware and Software
- Types of Programming Languages
- Compiler vs Interpreter
- Program Development Life Cycle
- Algorithm and Flowchart
- Structure of a C Program
- Installing Compiler (GCC/Turbo C/VS Code)

Lab

- First C Program
- Compile and Execute Program

Module 2: Basics of C Programming

Topics

- Character Set
- Tokens in C
- Keywords
- Identifiers
- Constants
- Variables
- Data Types
- Declaration and Initialization
- Input and Output Functions
    - printf()

  - scanf()

Lab Programs

- Print Name and Address
- Addition of Two Numbers
- Area of Circle
- Simple Interest

Module 3: Operators and Expressions

Topics

- Arithmetic Operators
- Relational Operators
- Logical Operators
- Assignment Operators
- Increment and Decrement Operators
- Conditional Operator
- Operator Precedence

Lab Programs

- Calculator Program
- Swap Two Numbers
- Largest of Two Numbers

Module 4: Decision Making Statements

Topics

- if Statement
- if-else Statement
- Nested if
- else-if Ladder
- switch Statement

Lab Programs

- Even or Odd
- Positive or Negative
- Largest of Three Numbers
- Grade Calculation
- Menu Driven Calculator

Module 5: Looping Statements

Topics

- while Loop
- do-while Loop
- for Loop
- Nested Loops
- break Statement
- continue Statement

Lab Programs

- Multiplication Table
- Factorial
- Fibonacci Series
- Prime Number
- Armstrong Number
- Reverse Number

Module 6: Arrays

Topics

- One Dimensional Array
- Two Dimensional Array
- Array Initialization
- Array Operations

Lab Programs

- Sum of Array Elements
- Largest Element
- Sorting Array
- Matrix Addition
- Matrix Multiplication

Module 7: Strings

Topics

- String Declaration
- String Input and Output
- String Handling Functions
    - strlen()
    - strcpy()
    - strcat()

- strcmp()

Lab Programs

- Reverse String
- Palindrome String
- Count Vowels
- String Concatenation

Module 8: Functions

Topics

- Introduction to Functions
- Function Declaration
- Function Definition
- Function Call
- Types of Functions
- Recursion

Lab Programs

- Factorial using Function
- Prime Number using Function
- Recursive Factorial
- Recursive Fibonacci

Module 9: Pointers

Topics

- Introduction to Pointers
- Pointer Variables
- Address Operator
- Dereferencing
- Pointer Arithmetic

Lab Programs

- Swap Using Pointers
- Array Using Pointers
- String Using Pointers

Module 10: Structures and Unions

Topics

- Structure Declaration

- Structure Variables
- Array of Structures
- Nested Structures
- Union Basics

Lab Programs

- Student Information System
- Employee Details

Module 11: File Handling

Topics

- Files in C
- fopen()
- fclose()
- fprintf()
- fscanf()
- fread()
- fwrite()

Lab Programs

- Student Record Management
- Employee Database

Module 12: Dynamic Memory Allocation

Topics

- malloc()
- calloc()
- realloc()
- free()

Lab Programs

- Dynamic Array Creation
- Student Records Storage

Module 13: Preprocessor Directives

Topics

- #include
- #define
- Macros

- Header Files
- Conditional Compilation

Lab Programs

- Macro Examples
- Custom Header File

Module 14: Searching and Sorting

Topics

- Linear Search
- Binary Search
- Bubble Sort
- Selection Sort

Lab Programs

- Search Student Record
- Sort Marks List

Module 15: Mini Projects

Project 1

Student Management System

Project 2

Library Management System

Project 3

Bank Account Management System

Project 4

Employee Payroll System

Learning Outcomes

After completing this course, students will be able to:

✅ Write C programs independently
✅ Solve programming problems using algorithms
✅ Implement arrays, strings, functions, pointers, and structures
✅ Perform file handling operations
✅ Develop small real-world applications
✅ Build a strong foundation for Data Structures, C++, Java, and Python

Recommended Lab Practice

- Minimum: 100 Programs

- Ideal: 150–200 Programs
- Mini Projects: 4
- Assignments: 10
- Weekly Assessments: 5

This syllabus is suitable for **B.Tech First Year Students, Diploma Students, and Campus Placement Foundation Programs**.